

Fixed-Complexity Cell Survival and Sharp Concentration Obstructions for the TPC Static Mask

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Abstract

The preceding twin-prime-correlation (TPC) reduction left a precise finite-dimensional question: does the literal static row mask retain enough of the squared mass of the residue/Mellin cells to compare a bounded-projective envelope with an atomic diagonal? We first remove a representation ambiguity. Weighted survival is canonically attached to a *pre-mask* cell—the fixed-period residue and real-axis Mellin factors are separated before the Fourier–Ferrers–gcd representation of the static mask is introduced. It is not an invariant of an arbitrary final projective layer.

For a binary mask M and a nonzero right weight b , let

$$\eta_M(b) = \max_{\alpha} \frac{\sum_{\gamma: M(\alpha, \gamma)=0} |b_{\gamma}|^2}{\|b\|_2^2}.$$

We prove the exact minimax identity

$$\inf_{a \neq 0} \delta_w(M; a, b) = 1 - \eta_M(b).$$

For the TPC mask, the necessary-and-sufficient quantity η_M is bounded by three explicit anti-concentration numbers measuring mass on a single prime-source fibre, in a determinant window of width $\Delta = QX^{-\kappa_0}$, and on a large-gcd fibre. These are the geometric sufficient quantities used below; they are individually sharp, whereas their sum need not be necessary because the deleted sets may overlap.

We then unpack the actual fixed-complexity provenance. An identified nondegenerate direct opposite-row cell has a fixed smooth dyadic shape, a fixed admissible residue set, and unit-modulus residue and real-axis Mellin phases. Fixed-modulus prime and squarefree counts give uniformly

$$\text{Flat}(b) = O(1).$$

Together with the complete-row degree theorem, this yields

$$\delta_w(M; a, b) = 1 - O\left(L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}\right) = 1 - o(1)$$

for every nonzero weight a on the other row, however sparse. Hence, on this identified separated row-output face, the TPC–36 representation has only $X^{o(1)}$ projective-to-atomic energy distortion: the active-cell survival gate consumes no polynomial part of the $1/400$ endpoint allowance. An abstract finite-family selection lemma gives the same survival conclusion for energetic cells under an explicit aggregate lower-frame hypothesis, while controlling their aggregate discarded cell energy. It is not a frame theorem for the signed continuous Mellin reassembly.

The distinction from the upper-only inherited interface is sharp. Concentration on a same-source fibre, a near-product block, or a large-gcd block produces polynomially large packets with zero survival; adding one admitted neighbor gives positive survival as small as the inverse packet size. Such selectors require growing or shrinking complexity and are not generated by the literal fixed-period, fixed-shape, real-axis Mellin provenance. The paper does not compare the full signed TPC-48 profile envelope with the physical atomic diagonal, prove a Mellin frame lower bound, control the coherent prime square, group residual labels, or localize the frozen family back to the physical shift. No parity-barrier breach or twin-prime conclusion is claimed.

Keywords: weighted mask survival; residue cells; Mellin separation; anti-concentration; projective tensor norm; determinant equations; sieve parity.

Convention. All finite weights may be complex; survival depends only on their moduli. Zero cells are deleted. Constants may depend on the fixed shift, fixed residue moduli, and fixed smooth shapes, but never on the real Mellin parameters or on X .

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1 Introduction

TPC-48 reduces the completed actual-coefficient determinant family to a signed projective row-square envelope, while TPC-49 identifies the exact finite-dimensional distortion between a Hilbertized projective envelope and its atomic energy [8, 9]. For a weighted mask matrix

$$T = D_a K D_b, \quad \delta_w(K; a, b) = \frac{\sum_{\alpha, \gamma} |a_\alpha|^2 |K_{\alpha\gamma}|^2 |b_\gamma|^2}{\|a\|_2^2 \|b\|_2^2}, \quad (1)$$

that comparison gives

$$\frac{\mathcal{E}_{\mathcal{R}}(T)^2}{\|T\|_{S_2}^2} \leq \frac{P_{\mathcal{R}}(K)^2}{\delta_w(K; a, b)}. \quad (2)$$

Here $P_{\mathcal{R}}$ is the bounded-projective cost of one specified mask representation. TPC-36 supplies a representation $\mathcal{R}_{36}$ with $P_{\mathcal{R}_{36}} = X^{o(1)}$ [5]. The unresolved quantity was therefore the weighted survival of the actual active cells.

The phrase “actual active cell” hides an order-of-operations issue. The final parameter ω in the TPC-47/48 representation already combines fixed-residue and Mellin separation with the Fourier–Ferrers–gcd decomposition of the static mask [7, 8]. It is not a canonical physical cell. Null pairs can be appended to an exact signed representation, so no nontrivial lower-mass statement can hold uniformly for every final projective layer. The static mask must instead be left intact while the fixed-period and smooth factors are separated. This produces a specified *pre-mask cell*; only then is (1) the right quantity.

1.1 The exact concentration criterion

For a binary mask M , define the largest forbidden proportion of a right weight $b \neq 0$ by

$$\eta_M(b) := \max_{\alpha} \frac{\sum_{\gamma: M(\alpha, \gamma)=0} |b_\gamma|^2}{\|b\|_2^2}. \quad (3)$$

Our first main theorem is the exact identity

$$\boxed{\inf_{a \neq 0} \delta_w(M; a, b) = 1 - \eta_M(b)}. \quad (4)$$

Thus a uniform statement over arbitrary pruning of the left row is equivalent to one-sided anti-concentration of the right weight. For the literal TPC static mask

$$M_X(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > QX^{-\kappa_0}} \mathbf{1}_{(d_\alpha, d_\gamma) \leq X^{\kappa_0}}, \quad (5)$$

the single quantity η_{M_X} is bounded by the sum of three transparent concentration coefficients: same-source, near-product, and large-gcd mass. This both refines the degree-flatness criterion of TPC-49 and identifies the minimal one-sided repair of the inherited upper-only interface.

1.2 Restoring fixed-complexity provenance

The physical opened row before static-mask decomposition has the form

$$\mu(d)(\log \ell)\omega_D(d)\psi_L(\ell/L)\zeta_{\ell,d}, \quad (6)$$

where the smooth shapes and residue periods are fixed [2, 3]. Finite residue Fourier characters and real-axis Mellin row factors have modulus one. The identified direct opposite-row factor uses $e = 0$; the restricted source-factor variant with its physical prime logarithm uses $e = 1$. Weighted prime and squarefree counts give, for either displayed weight class,

$$\|b\|_2^2 \asymp LD(\log L)^{2e-1}, \quad \|b\|_\infty^2 \ll (\log L)^{2e}, \quad e \in \{0, 1\}. \quad (7)$$

Since the complete row universe has $N \asymp LD/\log L$, this proves

$$\text{Flat}(b) := \frac{N\|b\|_\infty^2}{\|b\|_2^2} = O(1) \quad (8)$$

uniformly in every real Mellin parameter. Combining (8) with the complete-row degree theorem of TPC-49 proves

$$\boxed{\delta_w(M_X; a, b) = 1 - O\left(L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}\right)} \quad (9)$$

for every $a \neq 0$. In particular, arbitrary terminal sparsity on one side is harmless when the opposite side retains a canonical fixed-complexity cell.

At the inherited endpoint,

$$L = X^{99979/210000+o(1)}, \quad D = X^{10049/52500+o(1)}, \quad (10)$$

so (9) is $1 - o(1)$. In the isolated Hilbertized row-output face, its density exponent is therefore zero, rather than the maximal allowed value $1/400$. This closes the static survival subgate for the identified provenance-restored cells; it does not close the full TPC-48 envelope comparison.

1.3 Sharpness of the provenance requirement

The preceding conclusion cannot be recovered from row-count and norm upper bounds alone. The three deletion channels in (5) support correspondingly large concentrated packets with

$$\delta_w = 0. \quad (11)$$

A near-product block and a large-gcd block can have $X^{267/400-\kappa_0+o(1)}$ rows. Adding one admitted neighbor makes the masked matrix nonzero but leaves survival equal to the inverse of the packet size. At $\kappa_0 = 1/400$, this may be $X^{-133/200+o(1)}$, far below the isolated-face target.

These are exact logical counterexamples to the upper-only interface, not models of the literal coefficient. A same-source selector tracks one prime as X varies, a near-product selector shrinks at relative scale $X^{-\kappa_0}$, and a gcd selector uses a growing modulus. None is produced by a fixed-period residue expansion or by a unit-modulus real Mellin phase. The positive and negative theorems therefore meet at a clean boundary: fixed complexity gives survival, while growing concentration complexity must be estimated explicitly.

1.4 Scope and organization

The result proved here concerns one separated row-mask face. It does not identify the final TPC-48 parameter ω with a physical cell, replace a signed Mellin integral by an orthogonal frame, control the coherent prime sum at fixed residual coordinate, compare the full projective profile envelope with the literal atomic diagonal, group the map $s = d(m)r$, or recover the physical shift from the frozen family. These boundaries are structural, not disclaimers that can be removed by relabeling a layer.

Section 2 fixes the canonical order of separation. Section 3 proves the exact minimax criterion. Section 4 treats general subpower cell families. Section 5 proves the fixed-provenance theorem. Section 6 gives sharp concentration packets. Section 7 records the endpoint ledger and remaining gates, and section 8 states the exact finite certificate.

2 The canonical pre-mask cell interface

The weighted-survival parameter of TPC-49 is attached to a weighted two-row mask. It is therefore essential to define it before the static mask is expanded into signed rank-one layers. This section fixes that order of operations, records the nondegeneracy inherited from the physical packet, and states the exact transfer theorem available for one specified cell. In particular, a terminal projective parameter ω from TPC-47/48 is not declared to be a physical cell.

2.1 The physical coefficient before projective expansion

Fix the physical shift $h_0 \neq 0$ and one source-compatible dyadic packet. Write $\mathcal{A}_1$ and $\mathcal{A}_2$ for the two row systems. A row is

$$\rho = (\ell_\rho, d_\rho), \quad m_\rho = \ell_\rho d_\rho \asymp Q, \quad (12)$$

where ℓ_ρ is prime, d_ρ is squarefree, and the inherited separation makes m_ρ determine ρ . The literal static mask on $\mathcal{A}_1 \times \mathcal{A}_2$ is

$$K_{\alpha\gamma} := \mathbf{m}_{\text{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad (13)$$

with $\Delta = QX^{-\kappa_0}$ and $G = X^{\kappa_0}$. The joint multiplier is

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) = K_{\alpha\gamma} \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J). \quad (14)$$

Here Ξ is fixed-period in the orbit variable and $\mathcal{W}$ contains the inherited smooth factors. The physical source coefficient on the left has the form

$$\gamma_\alpha^{(1)} = \mu(d_\alpha) (\log \ell_\alpha) \omega_D^{(1)}(d_\alpha) \psi_L^{(1)}(\ell_\alpha/L) \zeta_\alpha^{(1)}, \quad (15)$$

and the right coefficient $\gamma_\gamma^{(2)}$ has the symmetric form. The inherited estimates are upper bounds only:

$$\sum_{\rho \in \mathcal{A}_s} |\gamma_\rho^{(s)}| \ll QX^{o(1)}, \quad \sum_{\rho \in \mathcal{A}_s} |\gamma_\rho^{(s)}|^2 \ll QX^{o(1)}, \quad s = 1, 2. \quad (16)$$

No lower row count, lower ℓ^2 -mass, or pointwise lower bound is part of this interface [5–8].

After large-prime descent, the literal left residual vector at h_0 is

$$\mathcal{F}_{h_0}^{\text{act}, L} = \sum_{\substack{\gamma, j, m, p, r \\ pr - mj = h_0}} \gamma_m^{(1)} \mathfrak{A}_{m, \gamma}^{\text{act}}(j) \mu(r) \Lambda(pr) e_{\gamma, d(m)r} \otimes e_j. \quad (17)$$

Thus the two axes relevant to static-mask survival are the source row m and the opposite row γ . The terminal factor, the Möbius phase, and the residual basis label are downstream data; they are not additional entries of the row mask.

2.2 Provenance-canonical residue/Mellin cells

We now fix one representation rather than taking an infimum over all algebraically equivalent ones. First split the physical orbit into the actual residue classes $a \pmod{H_0}$. On each class, apply the specific finite Fourier and Mellin inversions supplied by the inherited provenance construction, with their original normalizations and cutoffs. No zero pair, algebraic refactorization, or static-mask Fourier expansion is inserted at this stage. We call the resulting specified decomposition the *provenance-canonical pre-mask cellization* and denote it by $\mathcal{C}$.

Concretely, $\mathcal{C}$ is the exact identity

$$\begin{aligned} & \gamma_\alpha^{(1)} \Xi_{\alpha,\gamma}(j) \mathcal{W}_{\alpha,\gamma}(j/J) \\ &= \sum_{a \pmod{H_0}} \mathbf{1}_{j \equiv a \pmod{H_0}} \int_{\Theta_a} u_{a,\tau}^L(\alpha) v_{a,\tau}(\gamma) \phi_{a,\tau}(j/J) d\lambda_a(\tau). \end{aligned} \quad (18)$$

The source coefficient in (15) is absorbed into $u_{a,\tau}^L$. Equivalently, one may first separate $\Xi \mathcal{W}$ and then put $u_{a,\tau}^L(\alpha) = \gamma_\alpha^{(1)} u_{a,\tau}(\alpha)$. The right-side cellization is obtained by absorbing $\gamma_\gamma^{(2)}$ into the summation-row factor on the right.

There is one fixed compact set $I_0 \subset \mathbb{R}$, independent of X, a, τ , such that

$$\phi_{a,\tau} \in C_c^\infty(I_0). \quad (19)$$

The common cutoff used to obtain this support equals one throughout the physical normalized orbit range. Hence it does not alter a physical coefficient. For every fixed integer $B \geq 0$,

$$\sum_{a \pmod{H_0}} \int_{\Theta_a} \|u_{a,\tau}^L\|_\infty \|v_{a,\tau}\|_\infty \|\phi_{a,\tau}\|_{C^B} d|\lambda_a|(\tau) \ll_{B,h_0} X^{o(1)}. \quad (20)$$

This is an upper projective-seminorm estimate. It contains no lower bound for either unary factor. Identity (18) is precisely the point at which “cell” is defined in this paper: a cell is $c = (a, \tau)$ before K is decomposed.

Definition 2.1 (Literal pre-mask cell tensor). For $c = (a, \tau)$, put

$$T_c^L(\alpha, \gamma) := u_c^L(\alpha) K_{\alpha\gamma} v_c(\gamma). \quad (21)$$

The cell is called active when $T_c^L \neq 0$. Define its nonnegative unary weights by

$$a_c^L(\alpha) := |u_c^L(\alpha)|, \quad b_c(\gamma) := |v_c(\gamma)|. \quad (22)$$

The right tensor and weights are defined symmetrically.

Unary complex phases are harmless: diagonal unitary multiplication does not change Frobenius, nuclear, or projective-envelope norms. A genuine common nonzero scalar multiplies both numerator and denominator below and therefore cancels. A source-dependent terminal or Mellin modulus is instead absorbed into the weight a_c^L , while its unit phase still leaves survival unchanged.

2.3 The exact weighted-survival parameter

Definition 2.2 (Physical row-mask survival of a cell). For an active left cell define

$$\delta_c^L := \frac{\sum_{\alpha \in \mathcal{A}_1} \sum_{\gamma \in \mathcal{A}_2} a_c^L(\alpha)^2 K_{\alpha\gamma} b_c(\gamma)^2}{\left(\sum_{\alpha \in \mathcal{A}_1} a_c^L(\alpha)^2 \right) \left(\sum_{\gamma \in \mathcal{A}_2} b_c(\gamma)^2 \right)}. \quad (23)$$

Define δ_c^R by exchanging the two row systems.

Equivalently,

$$\delta_c^L = \frac{\|T_c^L\|_{S_2}^2}{\|a_c^L\|_2^2 \|b_c\|_2^2}. \quad (24)$$

Because K is a zero-one mask, an active cell satisfies only the qualitative inherited nondegeneracy

$$0 < \delta_c^L \leq 1. \quad (25)$$

There is no inherited lower bound by a power of X . In particular, neither (16) nor (20) implies $\delta_c^L \geq X^{-1/400+o(1)}$.

The failure is already visible on a complete finite mask. If one row has a forbidden neighbor γ_0 and an admitted neighbor γ_1 , take $a = e_\alpha$ and

$$b = (1 - \eta)^{1/2} e_{\gamma_0} + \eta^{1/2} e_{\gamma_1}, \quad 0 < \eta < 1. \quad (26)$$

Then $\delta_w(K; a, b) = \eta$. This is not asserted to be a physical TPC cell. It proves that upper norms, mask support, and bounded projective complexity alone logically imply no quantitative survival.

2.4 Why a terminal ω -layer has no physical survival parameter

The order in (18) is forced by the meaning of δ_c . After the pre-mask cell is fixed, TPC-36 expands

$$K_{\alpha\gamma} = \int_{\Omega_{36}} p_\eta(\alpha) q_\eta(\gamma) d\nu_{36}(\eta). \quad (27)$$

The final parameter used in TPC-47/48 combines c , η , and the terminal Mellin variables into one symbol ω . An individual such layer contains the rank-one factor

$$u_c^L(\alpha) p_\eta(\alpha), v_c(\gamma) q_\eta(\gamma), \quad (28)$$

not the weighted masked tensor $D_{ac} K D_{bc}$. The literal mask is recovered only after the complete signed η -integral is taken.

Consequently, assigning $\delta_w(K; |u_c p_\eta|, |v_c q_\eta|)$ to a final ω -layer would multiply by K a second time and measure a tensor different from the layer being reassembled. There are three further obstructions:

- (i) the signed projective representation is not unique;
- (ii) a cancelling null pair may be appended without changing the physical coefficient while changing every layerwise statistic;
- (iii) the TPC-48 layer also contains a profile seminorm, a coherent prime sum, and a residual Hilbert vector, so it is not a canonical two-axis matrix of the form $D_a K D_b$.

Thus no representation-independent “physical δ_ω ” is defined. The pointwise infimum in TPC-48 may remove inefficient representations at the level of the full envelope, but it does not turn their individual layers into physical atoms. Every survival assertion below is therefore made for the fixed pre-mask cellization $\mathcal{C}$ and the specified mask representation $\mathcal{R}_{36}$, never for all exact signed representations.

2.5 Specified $\mathcal{R}_{36}$ transfer

Let $\mathcal{R}_{36}$ be the explicit representation (27) furnished by same-source orthogonality, signed Ferrers separation, and global gcd compression. Its declared bounded-projective cost is

$$P_{36}(K) := \int_{\Omega_{36}} \|p_\eta\|_\infty \|q_\eta\|_\infty d|\nu_{36}|(\eta) \ll \log(2Q)(3 + 2\sqrt{2})^{\omega_*} = X^{o(1)}. \quad (29)$$

For an active left cell define

$$\mathcal{E}_{36,c}^L := \int_{\Omega_{36}} \|D_{a_c^L} p_\eta\|_2 \|D_{b_c} \overline{q_\eta}\|_2 d|\nu_{36}|(\eta). \quad (30)$$

Theorem 2.3 (Canonical-cell projective-to-atomic transfer). *Every active pre-mask cell satisfies*

$$\|T_c^L\|_{S_1} \leq \mathcal{E}_{36,c}^L \leq P_{36}(K)\|a_c^L\|_2\|b_c\|_2, \quad (31)$$

$$\boxed{\frac{(\mathcal{E}_{36,c}^L)^2}{\|T_c^L\|_{S_2}^2} \leq \frac{P_{36}(K)^2}{\delta_c^L}}. \quad (32)$$

The same formulas hold on the right.

Proof. Insert (27) into (21). After removing unary phases this is an exact Hilbert rank-one representation of $D_{a_c^L} K D_{b_c}$. The triangle inequality for the nuclear norm gives the first inequality in (31), while

$$\|D_{a_c^L} p_\eta\|_2 \|D_{b_c} \overline{q_\eta}\|_2 \leq \|a_c^L\|_2 \|b_c\|_2 \|p_\eta\|_\infty \|q_\eta\|_\infty \quad (33)$$

gives the second. Divide its square by (24). $\square$

In particular, if a new arithmetic theorem proves

$$\delta_c^L \geq X^{-\theta+o(1)}, \quad (34)$$

then the specified cell representation loses at most

$$\frac{(\mathcal{E}_{36,c}^L)^2}{\|T_c^L\|_{S_2}^2} \leq X^{\theta+o(1)}. \quad (35)$$

If the entire isolated coefficient-face allowance is assigned here, $\theta \leq 1/400$ is sufficient. This allocation is not a ledger for the complete TPC argument.

For comparison, on the complete-universe extension with N rows, TPC-49 proves minimum degree $(1 - \epsilon_X)N$, where

$$\epsilon_X \ll L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}. \quad (36)$$

Writing

$$\text{Flat}(b) := \frac{N\|b\|_\infty^2}{\|b\|_2^2}, \quad (37)$$

one then has the conditional bound

$$\delta_w(K; a, b) \geq 1 - \epsilon_X \text{Flat}(b). \quad (38)$$

Hence $\epsilon_X \text{Flat}(b_c) = o(1)$ would imply $\delta_c = 1 - o(1)$. The compressed TPC-45/48 upper-only interface does not identify its active row support with that complete universe and does not itself contain such a flatness estimate. In [proposition 5.1](#), we therefore return to the uncompressed TPC-25/36 pre-terminal provenance and identify the direct fixed-complexity opposite-row factors for which the flatness condition is a theorem. Any independent downstream pruning of that opposite row must still be checked through the concentration criterion of [theorem 3.1](#).

2.6 Boundary of the cell theorem

Theorem 2.3 is exactly a separated row-mask statement. It does not make any of the following identifications:

- (i) It does not compare the full TPC-48 Banach profile envelope $\mathfrak{S}_{\mathcal{R}}(\alpha)$ with the fixed-shift physical atomic diagonal $\mathcal{D}_0$. Signed $L^1(d|\lambda|)$ reassembly still needs a frame or no-cancellation theorem.

- (ii) It does not estimate the coherent prime square inside a fixed residual coordinate. The p -sum remains inside the norm.
- (iii) It does not control collisions or grouping under $s = d(m)r$. The mask face is formed before that map.
- (iv) It does not unfold sparse determinant aliases, localize the completed frozen family back to h_0 , or identify the residual Gram route with the native affine physical face.

Accordingly, even a subpower lower bound for every identified direct canonical cell would close only the corresponding active-cell survival subgate. It would not prove a fixed-shift Chowla estimate, cross the parity barrier, or imply any prime-pair or twin-prime conclusion.

3 Exact forbidden-mass concentration

The weighted-density condition of TPC-49 has an exact one-sided formulation. This formulation is useful here because the determinant and terminal restrictions may concentrate the source weight, whereas the opposite-row cell is the object whose provenance can still be tested directly.

Let I, J be finite nonempty sets. For an arbitrary kernel $K : I \times J \rightarrow \mathbb{C}$ and nonzero weights $a \in \mathbb{C}^I$, $b \in \mathbb{C}^J$, put

$$\delta_w(K; a, b) := \frac{\sum_{i \in I} \sum_{j \in J} |a_i|^2 |K(i, j)|^2 |b_j|^2}{\|a\|_2^2 \|b\|_2^2}. \quad (39)$$

For the exact concentration theorem below, specialize to a binary mask $M : I \times J \rightarrow \{0, 1\}$. We allow the numerator in (39) to be zero. Thus $\delta_w = 0$ when the product weight is supported entirely on forbidden entries. This is the continuous extension of the TPC-49 definition to a zero weighted matrix.

For a nonzero right weight b , define its forbidden-row concentration by

$$\eta_M(b) := \max_{i \in I} \frac{\sum_{j \in J: M(i, j)=0} |b_j|^2}{\|b\|_2^2}. \quad (40)$$

Theorem 3.1 (Exact one-sided survival criterion). *For every nonzero $b \in \mathbb{C}^J$,*

$$\boxed{\inf_{a \in \mathbb{C}^I \setminus \{0\}} \delta_w(M; a, b) = 1 - \eta_M(b).} \quad (41)$$

Consequently,

$$[\delta_w(M; a, b) \geq \delta_0 \text{ for every } a \neq 0] \iff \eta_M(b) \leq 1 - \delta_0 \quad (42)$$

for every $0 \leq \delta_0 \leq 1$.

Proof. For $i \in I$, set

$$s_b(i) := \frac{\sum_{j \in J} M(i, j) |b_j|^2}{\|b\|_2^2} = 1 - \frac{\sum_{j: M(i, j)=0} |b_j|^2}{\|b\|_2^2}. \quad (43)$$

Then

$$\delta_w(M; a, b) = \sum_{i \in I} \frac{|a_i|^2}{\|a\|_2^2} s_b(i). \quad (44)$$

The right side is a convex combination of the finitely many values $s_b(i)$. Its infimum is their minimum, attained by a point mass at any minimizing row. By (43), that minimum is $1 - \eta_M(b)$. This proves both assertions. $\square$

We now specialize the criterion to the three deletions in the TPC static mask. Let rows be $\alpha = (\ell_\alpha, d_\alpha)$, put $m_\alpha = \ell_\alpha d_\alpha$, and define

$$M_X(\alpha, \gamma) := \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}. \quad (45)$$

For a nonzero opposite-row weight b , introduce

$$\eta_{\text{src}}(b) := \max_{\ell} \frac{\sum_{\gamma: \ell_\gamma = \ell} |b_\gamma|^2}{\|b\|_2^2}, \quad (46)$$

$$\eta_{\text{near}}(b) := \sup_{x \in \mathbb{R}} \frac{\sum_{\gamma: |m_\gamma - x| \leq \Delta} |b_\gamma|^2}{\|b\|_2^2}, \quad (47)$$

$$\eta_{\text{gcd}}(b) := \max_d \frac{\sum_{\gamma: (d, d_\gamma) > G} |b_\gamma|^2}{\|b\|_2^2}. \quad (48)$$

The maxima range only over source and divisor labels that occur on the left row set. Enlarging those ranges can make the numerical upper bound larger, but it does not invalidate any displayed inequality.

Proposition 3.2 (Three-channel anti-concentration). *For the mask (45),*

$$\eta_{M_X}(b) \leq \eta_{\text{src}}(b) + \eta_{\text{near}}(b) + \eta_{\text{gcd}}(b). \quad (49)$$

Hence every nonzero source weight a satisfies

$$\boxed{\delta_w(M_X; a, b) \geq 1 - \eta_{\text{src}}(b) - \eta_{\text{near}}(b) - \eta_{\text{gcd}}(b)}. \quad (50)$$

In particular, the endpoint target $\delta_w \geq X^{-1/400+o(1)}$, uniformly in a , follows from

$$\eta_{\text{src}}(b) + \eta_{\text{near}}(b) + \eta_{\text{gcd}}(b) \leq 1 - X^{-1/400+o(1)}. \quad (51)$$

If the sum of the three concentrations is $o(1)$, then the stronger conclusion $\delta_w = 1 - o(1)$ holds uniformly in a .

Proof. For a fixed left row α , the forbidden opposite rows lie in the union

$$\{\gamma : \ell_\gamma = \ell_\alpha\} \cup \{\gamma : |m_\gamma - m_\alpha| \leq \Delta\} \cup \{\gamma : (d_\alpha, d_\gamma) > G\}. \quad (52)$$

The b^2 -mass of the three sets, divided by $\|b\|_2^2$, is at most the respective quantities in (46)–(48). Take the maximum over α and apply [theorem 3.1](#). $\square$

Remark 3.3 (Minimal versus decomposable conditions). The single quantity $\eta_{M_X}(b)$ is necessary and sufficient for uniform one-sided survival. The three channel quantities give a slightly stronger but geometrically interpretable sufficient condition. The examples in [section 6](#) show that each channel condition is individually sharp: none can be discarded under the inherited upper-only interface.

4 Energy-selected survival for a family of cells

The fixed-shape theorem below controls every nonzero standard cell. It is useful to have a second statement that remains valid when a partition contains boundary atoms with arbitrarily

small energy. The result is elementary, but it separates low cell energy from poor mask survival without selecting a favorable signed summand.

Let $M : I \times J \rightarrow \{0, 1\}$, where $|J| = N$, and suppose every row has at most ϵN forbidden entries. Let $b_\lambda \in \mathbb{C}^J$, $\lambda \in \Lambda$, be a finite cell family. Fix $B > 0$ and $A > 0$, and assume

$$|\Lambda| \leq S, \quad \|b_\lambda\|_\infty^2 \leq B^2, \quad \sum_{\lambda \in \Lambda} \|b_\lambda\|_2^2 \geq ANB^2. \quad (53)$$

The last inequality is only an aggregate lower frame bound. It does not assert orthogonality of the cells or a lower bound for their signed sum.

For $0 < \tau < 1$, call a cell energetic if

$$\|b_\lambda\|_2^2 \geq \tau NB^2. \quad (54)$$

Theorem 4.1 (Energy selection without a cellwise lower bound). *Every energetic cell and every nonzero $a \in \mathbb{C}^I$ satisfy*

$$\delta_w(M; a, b_\lambda) \geq 1 - \frac{\epsilon}{\tau}. \quad (55)$$

Moreover, the total energy of the nonenergetic cells is bounded by

$$\frac{\sum_{\lambda \text{ nonenergetic}} \|b_\lambda\|_2^2}{\sum_{\lambda} \|b_\lambda\|_2^2} \leq \frac{S\tau}{A}. \quad (56)$$

Consequently, if $\epsilon = X^{-c+o(1)}$, $S/A = X^{o(1)}$, and $\tau = \epsilon^{1/2}$, then every energetic cell has survival at least $1 - X^{-c/2+o(1)}$, while the nonenergetic cells carry only $X^{-c/2+o(1)}$ of the aggregate cell energy.

Proof. For an energetic cell,

$$\text{Flat}(b_\lambda) = \frac{N\|b_\lambda\|_\infty^2}{\|b_\lambda\|_2^2} \leq \frac{1}{\tau}. \quad (57)$$

The degree–flatness inequality gives (55). Each nonenergetic cell has energy less than τNB^2 ; there are at most S of them. Divide their sum by the aggregate lower bound in (53) to obtain (56). The last assertion is immediate. $\square$

Corollary 4.2 (Bounded-overlap partitions). *Suppose functions $w_\lambda : J \rightarrow \mathbb{C}$ have overlap at most R , satisfy*

$$\sum_{\lambda} w_\lambda(j) = 1 \quad (j \in J), \quad |w_\lambda(j)| \leq C, \quad (58)$$

and let $0 \neq g \in \mathbb{C}^J$ obey

$$\|g\|_2^2 \geq cN\|g\|_\infty^2. \quad (59)$$

Then $b_\lambda = gw_\lambda$ satisfies (53) with $S = |\Lambda|$, $B = C\|g\|_\infty$, and $A = c/(RC^2)$ (or any smaller positive value).

Proof. At every j , Cauchy–Schwarz and the overlap bound give

$$1 = \left| \sum_{\lambda} w_\lambda(j) \right|^2 \leq R \sum_{\lambda} |w_\lambda(j)|^2. \quad (60)$$

Multiply by $|g_j|^2$ and sum over j . Thus

$$\sum_{\lambda} \|gw_\lambda\|_2^2 \geq R^{-1} \|g\|_2^2 \geq (c/R)N\|g\|_\infty^2 = \frac{c}{RC^2}NB^2.$$

The pointwise upper bound is immediate. $\square$

TPC-18 uses a fixed bounded-overlap dyadic partition in the opened d -variable [2]; fixed residue cells have overlap one. Provided the base-flatness and aggregate lower-energy hypotheses in [corollary 4.2](#) are verified for the family at hand, [theorem 4.1](#) then protects its energy-significant boundary atoms even if a uniform lower bound is not imposed on every individual cutoff. It is deliberately an aggregate cell-energy statement. Turning that aggregate into the squared norm of a signed Mellin reassembly is a separate frame/no-cancellation problem.

5 Identified fixed-complexity physical cells survive the static mask

This section works *before* the projective decomposition of the static mask. This order is essential. The fixed residue selectors and the smooth row-orbit factors are first put into canonical cells, while the complete static mask remains one kernel. The factors occurring in a particular projective representation of that kernel are not absorbed into the row weights.

5.1 Canonical pre-mask row cells

Set $H = \text{rad } |h_0|$, and fix an integer $q = O_{h_0}(1)$. We take q to be a common multiple of all fixed row and orbit residue moduli in the packet, including the period H_0 used in (18), and let

$$\mathcal{R}_X := \left\{ (\ell, d) : L < \ell \leq 2L, \ell \text{ prime}, D < d \leq 2D, \mu^2(d) = 1, (d, H) = 1 \right\}. \quad (61)$$

We assume $L > 4D$, $L, D \rightarrow \infty$, and $L, D \leq X^C$ for one fixed C . Put $N_X = |\mathcal{R}_X|$. Then

$$N_X \asymp_H \frac{LD}{\log L}. \quad (62)$$

This follows from the prime number theorem and the elementary squarefree count with fixed coprimality conditions [1].

Let $\mathcal{A} \subset (\mathbb{Z}/q\mathbb{Z})^2$ be a fixed row-residue cell. We call it *admissible* when it contains a residue pair for which the prime progression and the squarefree progression in (61) both have positive local density. This is a finite, explicitly checkable condition. Let $\psi, \omega \in C^\infty([1, 2])$ be fixed nonzero shapes with positive squared integrals. This includes the constant shape one; compactly supported interior shapes are also allowed. Assume that $\mathcal{A}$ contains at least one locally admissible residue pair on which the two shapes contribute positive density. For $e \in \{0, 1\}$, real t_1, t_2 , a fixed unit-modulus periodic phase χ , and a scalar $c \neq 0$, define

$$b_{\ell, d} := c(\log \ell)^e \psi(\ell/L) \omega(d/D) \mathbf{1}_{(\ell, d) \bmod q \in \mathcal{A}} \chi(\ell, d) \ell^{it_1} d^{it_2}. \quad (63)$$

The case $e = 1$ includes a physical prime-source logarithm; the case $e = 0$ includes an unweighted opposite-row cell. Additional real-axis Mellin phases or fixed Fourier characters may be included in $\chi(\ell, d) \ell^{it_1} d^{it_2}$.

In the inherited opened-row notation, let $V = R_{\text{src}}^{1/2+o(1)}$ be the hard upper cutoff on d [3]. We call a pre-terminal dyadic block *nondegenerate interior* when, after one of finitely many fixed subdivisions and a rescaling of L, D , its full divisor interval obeys

$$2D \leq V \quad (64)$$

for all sufficiently large X , and its fixed unary shapes and admissible residue support have positive squared density as specified above. Thus the phrase is a quantitative support condition, not merely a name for a nonzero summand.

Proposition 5.1 (Identification of the direct opposite-row factors). *Let $\mathcal{C}_{\text{prov}} := \mathcal{C}$ be the pre-mask cellization obtained by applying exactly the fixed-period Fourier expansion and the real-axis Mellin inversion in the TPC-25 provenance formula, before applying any TPC-36 Fourier–Ferrers–gcd representation of the static mask. Every nonzero coefficient-free opposite-row factor on a nondegenerate interior pre-terminal block of TPC-25/36, extended by zero to $\mathcal{R}_X$, is of the form (63) with $e = 0$, up to replacing $\mathcal{A}$ by a fixed finite union of admissible residue pairs and making one of finitely many fixed dyadic subdivisions of the inherited row interval. If a source factor contains the physical logarithm and its remaining fixed residue coefficient is split into nonzero fixed values on admissible cells, the same statement holds with $e = 1$. A general bounded coefficient satisfying only $|\zeta_{\ell,d}| \ll 1$ is not covered by the $e = 1$ assertion and is instead retained in the arbitrary weight on the other side.*

Proof. Before separation, the inherited unary row coefficient is a product of $\mu(d)$, one possible factor $\log \ell$, fixed smooth dyadic cutoffs in ℓ/L and d/D , and a fixed-period residue factor [3, 5]. In the direct-cell implementation, residue splitting gives a fixed set $\mathcal{A}$ and one may take $\chi = 1$. Equivalently, in the finite Fourier implementation, each summand carries one fixed character χ , whose modulus is one on its admissible support. For the smooth row-orbit factors, real-axis Mellin inversion replaces each occurrence of $W(\ell dj/X)$ by a scalar transform coefficient, an orbit phase, and row powers

$$(\ell d)^{it} d^{is} = \ell^{it} d^{i(t+s)}.$$

These row powers have modulus one. Products of the finitely many fixed smooth shapes remain fixed smooth shapes, and all transform coefficients may be placed in the nonzero scalar c . This is exactly (63); the Möbius sign is another unit phase on the squarefree support.

On a nondegenerate interior block, the zero extension is coefficient-preserving because $\mathcal{R}_X$ is the complete prime-squarefree dyadic row universe for this pre-terminal packet: condition (64) removes the inherited hard cutoff $d \leq V$, while the fixed subdivision and rescaling align the two dyadic intervals. The omitted rows are therefore precisely zeros of its fixed unary supports. No static-mask representation factor p_η or q_η has yet been introduced. An independently pruned row subset created downstream is not covered by this identification. Any block failing (64) is outside the proposition, whether its truncated support has positive or vanishing relative density. Only after a separate recellization and verification of the fixed-shape or aggregate lower-energy hypotheses may theorem 4.1 select its energetic cells; that theorem does not by itself control their signed Mellin reassembly. $\square$

Corollary 5.2 (Location of downstream pruning). *In the left physical residual route, fix a direct pre-mask residue and Mellin layer and fixed downstream data (j, p, r) . The modulus of the unary factor indexed by the opposite row γ , namely $|v_{a,\tau}(\gamma)|$, is the $e = 0$ weight identified in proposition 5.1. The target squarefreeness, terminal weight, prime/cofactor restrictions, and source coefficient depend on the summation row m and downstream variables, and may therefore be absorbed into the arbitrary weight a . The right route is symmetric.*

Proof. Inspect (17): after the fixed factorization of $\Xi\mathcal{W}$, the terminal and target data multiply the source integer m , whereas γ remains the opposite output coordinate and carries only its separated pre-mask unary factor. This is a per-layer statement; it does not identify the signed reassembled Mellin envelope with the physical atomic diagonal, nor may the absorption be moved through the coherent downstream prime sum without a new argument. $\square$

Theorem 5.3 (Fixed-complexity row energy). *For every fixed admissible cell as above, there are constants $\mathbf{c}_{\text{cell}}, F_{\text{cell}} \in (0, \infty)$ such that*

$$\sum_{(\ell,d) \in \mathcal{R}_X} |b_{\ell,d}|^2 = (\mathbf{c}_{\text{cell}} + o(1)) |c|^2 L D (\log L)^{2e-1}. \quad (65)$$

Moreover,

$$\|b\|_\infty^2 \ll_{\text{cell}} |c|^2 (\log L)^{2e}. \quad (66)$$

Both statements are uniform in all real Mellin parameters and all unit-modulus phases in (63). Consequently

$$\boxed{\text{Flat}_{\mathcal{R}_X}(b) := \frac{N_X \|b\|_\infty^2}{\|b\|_2^2} \leq F_{\text{cell}}.} \quad (67)$$

Proof. Split the fixed set $\mathcal{A}$ into its finitely many residue pairs. For every prime-admissible class, the fixed-modulus weighted prime number theorem gives [1]

$$\sum_{\substack{L < \ell \leq 2L \\ \ell \text{ prime} \\ \ell \equiv a \pmod{q}}} (\log \ell)^{2e} |\psi(\ell/L)|^2 = (c_{a,\psi} + o(1)) L (\log L)^{2e-1}, \quad (68)$$

where $c_{a,\psi} \geq 0$. Likewise, Möbius inversion followed by counting in a fixed progression gives

$$\sum_{\substack{D < d \leq 2D \\ d \equiv b \pmod{q} \\ \mu^2(d)=1, (d,H)=1}} |\omega(d/D)|^2 = (c_{b,\omega,H} + o(1)) D, \quad (69)$$

with a nonnegative, explicitly determined local-density constant. Admissibility and nontriviality of the fixed shapes make the sum of the products $c_{a,\psi} c_{b,\omega,H}$ over $(a,b) \in \mathcal{A}$ strictly positive. This proves (65). Boundedness of the fixed shapes gives (66). The Mellin and Fourier factors have modulus one, so none of these estimates depends on their real parameters. Finally combine (62), (65), and (66). $\square$

The hypothesis is not an unproved distribution conjecture. For every named fixed smooth and residue cell it is decided by a finite local check, and the two asymptotics in the proof are standard fixed-modulus theorems. A zero or locally forbidden Fourier term is simply omitted.

5.2 Weighted survival with arbitrary pruning on the other side

On $\mathcal{R}_X \times \mathcal{R}_X$, put

$$M_X(\alpha, \gamma) := \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad (70)$$

$$m_\alpha := \ell_\alpha d_\alpha, \quad \Delta := LDX^{-\kappa_0}, \quad G := X^{\kappa_0}, \quad (71)$$

where $\kappa_0 > 0$ is fixed. The complete-universe degree theorem of TPC-49 [9] gives, uniformly in every row on either side,

$$\#\{\gamma : M_X(\alpha, \gamma) = 0\} \leq \epsilon_X N_X, \quad \epsilon_X \ll_H L^{-1+o(1)} + D^{-1+o(1)} + X^{-\kappa_0+o(1)}. \quad (72)$$

For nonnegative weights a, b , with neither identically zero, define

$$\delta_w(M_X; a, b) := \frac{\sum_{\alpha, \gamma} a_\alpha^2 M_X(\alpha, \gamma) b_\gamma^2}{(\sum_\alpha a_\alpha^2) (\sum_\gamma b_\gamma^2)}. \quad (73)$$

Theorem 5.4 (Canonical-cell survival). *Let b be the modulus of a nonzero direct provenance factor from proposition 5.1; or, more generally, let it be the modulus of a canonical cell weight (63). For every nonzero weight a on the other row system, with completely arbitrary support and concentration,*

$$\boxed{\delta_w(M_X; a, b) \geq 1 - F_{\text{cell}} \epsilon_X = 1 - o(1).} \quad (74)$$

The conclusion is uniform in the real Mellin parameters and fixed unit-modulus residue phases. It also holds after exchanging the two row systems.

Proof. For every fixed α , (72) and the definition of the supremum norm give

$$\begin{aligned} \sum_{\gamma} M_X(\alpha, \gamma) b_{\gamma}^2 &\geq \|b\|_2^2 - \epsilon_X N_X \|b\|_{\infty}^2 \\ &\geq (1 - F_{\text{cell}} \epsilon_X) \|b\|_2^2, \end{aligned} \quad (75)$$

where the second line is (67). Multiply by a_{α}^2 , sum over α , and divide by the denominator in (73). No lower-density information about a is used. $\square$

This asymmetric formulation is useful for the physical residual packet. The arbitrary side may carry the squarefree target selector, large-prime/cofactor restrictions, terminal weights, or any other coefficient-specific pruning. Only the opposite row cell needs the fixed-complexity flatness proved above. If a later reduction also independently prunes that opposite row, the theorem is not automatic: one must recheck the concentration quantities in (46)–(48).

5.3 Transfer through the specified TPC–36 representation

Let $\mathcal{R}_{36}$ be the specified representation obtained from same-source Fourier orthogonality, signed Ferrers separation, and the global gcd compression. As proved in TPC–36 and transferred through the weighted interface in TPC–49, its bounded-projective mass satisfies [5, 9]

$$P_{\mathcal{R}_{36}}(M_X) = X^{o(1)}. \quad (76)$$

Write $\mathcal{E}_{\mathcal{R}_{36}}(M_X; a, b)$ for its weighted Hilbert envelope. The fixed-representation weighted transfer inequality is

$$\frac{\mathcal{E}_{\mathcal{R}_{36}}(M_X; a, b)^2}{\|D_a M_X D_b\|_{S_2}^2} \leq \frac{P_{\mathcal{R}_{36}}(M_X)^2}{\delta_w(M_X; a, b)}. \quad (77)$$

Corollary 5.5 (Subpower atomic cost on an identified canonical cell). *Under the hypotheses of theorem 5.4,*

$$\boxed{\mathcal{E}_{\mathcal{R}_{36}}(M_X; a, b)^2 \leq X^{o(1)} \|D_a M_X D_b\|_{S_2}^2.} \quad (78)$$

The estimate is uniform in the canonical real Mellin and fixed residue parameters.

Proof. Insert (74) and (76) into (77). $\square$

The disjoint orbit classes $j \bmod H_0$ are orthogonal output subspaces and therefore reassemble by an exact square sum. Disjoint source-row residue cells reassemble exactly only at the atomic Frobenius-diagonal level; after source summation they may contribute coherently to the same physical output coordinate. A finite character expansion has an $O_H(1)$ upper triangle/Bessel cost; Parseval is exact only if its auxiliary residue label is retained in an enlarged space. Real Mellin phases preserve (67), and every polynomially weighted variation of a fixed smooth Mellin transform is finite. These facts give a constant or $X^{o(1)}$ upper reassembly cost. They do not assert that distinct continuous Mellin layers are mutually orthogonal or supply a lower frame bound for the signed physical coefficient.

Proposition 5.6 (Why the older upper-bound interface is insufficient). *If the periodic/phase provenance in (63) is replaced only by an arbitrary coefficient $\zeta_{\ell,d}$ satisfying $|\zeta_{\ell,d}| \leq 1$, then no positive uniform lower bound for δ_w follows from the inherited upper norm bounds alone.*

Proof. Fix a row $\alpha_0 = (\ell_0, d_0)$, put a on that row alone, and put b on any nonempty subset of the opposite rows with source prime ℓ_0 . The same-source factor in (70) vanishes on every pair in the resulting rectangle, so $D_a M_X D_b = 0$ and $\delta_w = 0$. The weights remain bounded and obey all purely upper ℓ^1 , ℓ^2 , and support estimates. What excludes this example in theorem 5.3 is the unpacked fixed-complexity smooth/residue provenance, not an upper norm bound. $\square$

6 Sharp concentration obstructions

The exact criterion of [theorem 3.1](#) cannot be deduced from row-count and coefficient upper bounds. We give one large zero-survival block for each deletion in the literal static mask, and then remove the apparent degeneracy caused by a zero weighted matrix.

Fix $H \geq 1$, let $L, D \rightarrow \infty$, and assume $L > 4D$. Work on the complete dyadic row universe

$$\mathcal{R}_X := \left\{ (\ell, d) : L < \ell \leq 2L, \ell \in \mathbb{P}, D < d \leq 2D, \mu^2(d) = 1, (d, H) = 1 \right\}. \quad (79)$$

Put

$$N := |\mathcal{R}_X| \asymp_H \frac{LD}{\log L}, \quad Q := LD, \quad \Delta := QX^{-\kappa_0}, \quad G := X^{\kappa_0}, \quad (80)$$

where $\kappa_0 > 0$ is fixed. We assume $\Delta \rightarrow \infty$ and, only for the gcd construction, $G = o(D)$. Let M_X be the mask in [\(45\)](#) on $\mathcal{R}_X \times \mathcal{R}_X$.

6.1 One sharp block for each deleted channel

Proposition 6.1 (Same-source zero-survival block). *For all sufficiently large X , there is a prime $\ell_0 \in (L, 2L]$ such that*

$$S_{\text{src}} := \{(\ell_0, d) \in \mathcal{R}_X\} \quad (81)$$

satisfies

$$|S_{\text{src}}| \asymp_H D, \quad M_X(\alpha, \gamma) = 0 \quad (\alpha, \gamma \in S_{\text{src}}). \quad (82)$$

Consequently

$$\delta_w(M_X; \mathbf{1}_{S_{\text{src}}}, \mathbf{1}_{S_{\text{src}}}) = 0. \quad (83)$$

At the inherited endpoint,

$$|S_{\text{src}}| = X^{10049/52500+o(1)}. \quad (84)$$

Proof. The prime number theorem supplies ℓ_0 , and the elementary squarefree count with the fixed coprimality condition gives $|S_{\text{src}}| \asymp_H D$. Every pair in the block has equal source prime, so the first factor in [\(45\)](#) vanishes. The endpoint size is the inherited value of D . $\square$

Proposition 6.2 (Near-row zero-survival block). *There is a set $S_{\text{near}} \subset \mathcal{R}_X$ such that*

$$|S_{\text{near}}| \gg NX^{-\kappa_0}, \quad |m_\alpha - m_\gamma| \leq \Delta \quad (\alpha, \gamma \in S_{\text{near}}). \quad (85)$$

Therefore

$$\delta_w(M_X; \mathbf{1}_{S_{\text{near}}}, \mathbf{1}_{S_{\text{near}}}) = 0. \quad (86)$$

At the inherited endpoint,

$$|S_{\text{near}}| \geq X^{267/400-\kappa_0+o(1)}. \quad (87)$$

In particular, if $\kappa_0 = 1/400$, this is $X^{133/200+o(1)}$.

Proof. Every row product lies in $(Q, 4Q]$. Partition that interval into half-open intervals of length at most Δ . The number of intervals is

$$\left\lceil \frac{3Q}{\Delta} \right\rceil \ll X^{\kappa_0}. \quad (88)$$

By pigeonhole, one interval contains at least $\gg NX^{-\kappa_0}$ row products. Take the corresponding rows for S_{near} . Two products in the same half-open interval differ by less than Δ , so the middle factor in [\(45\)](#) vanishes. Finally $N = X^{267/400+o(1)}$ at the endpoint. $\square$

Proposition 6.3 (Large-gcd zero-survival block). *Assume $G = o(D)$. For all sufficiently large X , there is a prime g satisfying*

$$G < g \leq 2G, \quad (g, H) = 1, \quad (89)$$

and a set

$$S_{\text{gcd}} := \{(\ell, ge) \in \mathcal{R}_X : D/g < e \leq 2D/g, \mu^2(e) = 1, (e, gH) = 1\} \quad (90)$$

such that

$$|S_{\text{gcd}}| \asymp_H \frac{N}{g} = NX^{-\kappa_0+o(1)}, \quad M_X(\alpha, \gamma) = 0 \quad (\alpha, \gamma \in S_{\text{gcd}}). \quad (91)$$

Thus

$$\delta_w(M_X; \mathbf{1}_{S_{\text{gcd}}}, \mathbf{1}_{S_{\text{gcd}}}) = 0. \quad (92)$$

At the inherited endpoint the block has size

$$|S_{\text{gcd}}| = X^{267/400-\kappa_0+o(1)}. \quad (93)$$

Proof. The prime number theorem in the interval $(G, 2G]$, with the finitely many prime divisors of H deleted, gives g . Since $D/g \rightarrow \infty$, the squarefree count with the fixed coprimality restrictions gives $\asymp_H D/g$ choices of e . Multiplication by the $\asymp L/\log L$ source primes proves the cardinality statement. For any two rows in the block, both divisor coordinates are divisible by $g > G$. The last factor in (45) therefore vanishes. $\square$

The three constructions saturate the degree-flatness mechanism channel by channel. The source block has relative size $D/N = L^{-1+o(1)}$, while the near and gcd blocks have relative size at least $X^{-\kappa_0+o(1)}$; they may be thinned to that order while retaining zero survival. Their full-universe flatness factors are the reciprocals of exactly those quantities.

6.2 Nonzero packets with polynomially small survival

The preceding examples have zero weighted matrices. The next lemma shows that their obstruction is not confined to that endpoint.

Lemma 6.4 (One admitted neighbor). *Let $S \subset \mathcal{R}_X$ be any of $S_{\text{src}}, S_{\text{near}}, S_{\text{gcd}}$, and fix $\alpha_0 \in S$. For all sufficiently large X , there exists $x \in \mathcal{R}_X \setminus S$ with*

$$M_X(\alpha_0, x) = 1. \quad (94)$$

For

$$a = e_{\alpha_0}, \quad b = \mathbf{1}_S + e_x, \quad (95)$$

one has the exact value

$$\delta_w(M_X; a, b) = \frac{1}{|S| + 1}. \quad (96)$$

Proof. TPC-49 proves that every row of the complete static-mask graph has degree $(1 - o(1))N$ [9]. Hence α_0 has an admitted neighbor for sufficiently large X . Every pair in $\{\alpha_0\} \times S$ is forbidden by the defining property of the chosen block, so such a neighbor necessarily lies outside S . For (95), the numerator of (39) is exactly one, while $\|a\|_2^2 = 1$ and $\|b\|_2^2 = |S| + 1$. $\square$

Corollary 6.5 (Endpoint separation from the required density). *The same-source construction gives a nonzero weighted matrix with*

$$\delta_w = X^{-10049/52500+o(1)}. \quad (97)$$

The near and gcd constructions give nonzero weighted matrices with

$$\delta_w \leq X^{-(267/400-\kappa_0)+o(1)}. \quad (98)$$

When $\kappa_0 = 1/400$, the latter exponent is $133/200$, and all of these displayed survival parameters are polynomially smaller than the isolated TPC target $X^{-1/400+o(1)}$.

6.3 Why fixed provenance excludes these selectors

Proposition 6.6 (Unit-phase immunity). *Let $u : I \rightarrow \mathbb{C}$, $v : J \rightarrow \mathbb{C}$ have unit modulus and let $U : I \times J \rightarrow \mathbb{C}$ have unit modulus on the support of M . Then*

$$\delta_w(M \odot U; u \odot a, v \odot b) = \delta_w(M; a, b). \quad (99)$$

In particular, row and orbit Mellin phases, Möbius signs, and fixed residue characters cannot reduce weighted survival. A scalar Mellin coefficient cancels from the normalized ratio when it is nonzero; a zero scalar cell may be deleted.

Proof. Every term in (39) depends only on the absolute values of the two unary weights and of the kernel entry. $\square$

Remark 6.7 (Upper-only compatibility and physical incompatibility). The weights in [propositions 6.1 to 6.3](#) and [lemma 6.4](#) are bounded by one, have support at most N , and obey all row-count, pointwise, and ℓ^2 upper bounds inherited by TPC-45/48. They are therefore genuine countermodels to any deduction based only on that upper interface. They are not asserted to be the literal physical coefficient.

More precisely, the same-source block selects one prime that moves with X ; the near block requires a normalized cutoff of width $X^{-\kappa_0}$; and the gcd block requires divisibility by a modulus $g \asymp X^{\kappa_0}$. None is generated by a nonzero fixed smooth dyadic family, a fixed-period residue cell, or a unit-modulus Mellin phase. A power-narrow smooth selector would also violate the inherited uniform normalized C^B -bounds. Thus the examples prove the need for an explicit no-concentration or fixed-provenance theorem; they do not disprove survival for the literal fixed-complexity cells.

Finally, cancellation between different signed residue/Mellin layers does not alter this conclusion. Such cancellation changes the representation condition number and belongs to the frame-efficiency gate, not to the nonnegative survival ratio (39).

7 Endpoint ledger and the boundary of the result

7.1 The isolated active-cell exponent

At the inherited endpoint, write

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad QJ = X^{1+o(1)}, \quad (100)$$

and use the finer row factorization

$$L = X^{99979/210000+o(1)}, \quad D = X^{10049/52500+o(1)}. \quad (101)$$

The opened-divisor cutoff is genuinely inactive here. Indeed, the high- β source packet of TPC-28 has [4]

$$R_{\text{src}} = X^{23/60+o(1)}, \quad V = R_{\text{src}}^{1/2+o(1)} = X^{23/120+o(1)}, \quad (102)$$

and

$$\frac{23}{120} - \frac{10049}{52500} = \frac{9}{35000} > 0. \quad (103)$$

Consequently $2D \leq V$ for all sufficiently large X , with a fixed power margin. Thus the actual high- β packet contains the nondegenerate interior dyadic row blocks required in [proposition 5.1](#); this is not an empty support hypothesis. Thus (72) becomes

$$\epsilon_X = O\left(X^{-\eta+o(1)}\right), \quad \eta := \min\left\{\kappa_0, \frac{10049}{52500}\right\} > 0. \quad (104)$$

The D^{-1} term dominates the L^{-1} term unless the fixed pruning exponent κ_0 is smaller. In particular,

$$\delta_w(M_X; a, b) = 1 - O\left(X^{-\eta+o(1)}\right) \quad (105)$$

for every fixed identified nondegenerate direct opposite-row cell and every arbitrary nonzero weight on the other side. The implied constant may depend on that fixed cell.

Recall the isolated-face budget notation

$$P_{\mathcal{R}} = X^{\varpi+o(1)}, \quad \delta_w = X^{-\theta+o(1)}. \quad (106)$$

The fixed-representation sufficient condition at this endpoint is

$$2\varpi + \theta \leq \frac{1}{400}. \quad (107)$$

Corollary 7.1 (Zero polynomial cost for the canonical static face). *For the representation $\mathcal{R}_{36}$ and every identified nondegenerate direct opposite-row pre-mask cell,*

$$\varpi = 0, \quad \theta = 0. \quad (108)$$

Consequently this isolated static-mask face consumes no positive polynomial portion of the $X^{1/400+o(1)}$ allowance.

Proof. Equation (76) gives $\varpi = 0$, and (105) gives $\theta = 0$. Substitute in (107). $\square$

The phrase *consumes no polynomial allowance* concerns exactly the Hilbertized static row-output face in (78). It is not a statement that the remaining $1/400$ allowance closes the full determinant packet. If all remaining operations have total loss exponent λ_{other} , the full multiplicative ledger would still require

$$\lambda_{\text{other}} \leq \frac{1}{400}. \quad (109)$$

No estimate for λ_{other} is inferred here.

7.2 What has and has not been proved

The positive theorem removes one precise obstruction. Even when the source side is arithmetically sparse, the complete static mask deletes only $o(1)$ of the product ℓ^2 -mass of every identified fixed-complexity opposite-row residue/Mellin cell covered by [proposition 5.1](#). Therefore active-row sparsity by itself does not create a polynomial projective-to-atomic loss in this canonical order of decomposition.

The following interfaces remain open and are not consequences of [theorem 5.4](#) and [corollary 5.5](#).

- (i) **Continuous Mellin frame efficiency.** The real Mellin phases preserve row flatness, and Schwartz variation gives finite weighted L^1 mass, but two distinct Mellin layers are not automatically orthogonal in the physical coefficient Hilbert space. Minkowski reassembly supplies an upper envelope; it does not supply a lower frame bound comparing that envelope with the squared norm of the signed, reassembled physical coefficient. Selecting one favorable Mellin parameter by pigeonhole is invalid.
- (ii) **The full TPC-48 Banach profile envelope.** The quantity controlled in (78) is one genuinely Hilbertized row-output face. The TPC-48 envelope also carries the large-sieve profile constant and an outer signed Bochner integral. This paper does not identify that Banach profile envelope with a nuclear norm or with the physical diagonal.

- (iii) **The coherent prime square.** At fixed source m and cofactor r , the prime sum is formed before taking the norm. The exact orthogonality between distinct residual labels separates r , but it does not separate the primes inside

$$\sum_p A_\omega(p, r, m) e(-pr\alpha) v_{\omega, p, r, m}. \quad (110)$$

No prime exponential-sum cancellation, dispersion estimate, or pointwise large-sieve replacement is proved here.

- (iv) **Residual grouping.** The residual lift distinguishes the labels $s = d(m)r$. Returning to the grouped physical output applies a map whose squared norm is the maximal residual-fiber multiplicity. The present row-mask theorem neither bounds that multiplicity by $X^{o(1)}$ nor finds cancellation between labels in one fiber.
- (v) **Frozen completion versus the physical shift.** The frozen family extends the coefficient field constructed at h_0 to determinant equations with arbitrary intercept h . Only $h = h_0$ has physical provenance. A square estimate for the completed family need not have the same normalization as the one-slice atomic diagonal. No completed-to- h_0 localization theorem is supplied.
- (vi) **Residual and native affine Gram faces.** The large-prime residual synthesis is obtained after low-sign averaging. Its cross-inner-product structure is not automatically the native affine physical Gram structure whose atomic diagonal is $\mathcal{D}_0$. The static survival theorem does not identify those two faces.
- (vii) **Slope collision and the remaining endpoint ledger.** The common-profile large sieve may still incur determinant-slope collision, and none of that cost is cancelled by (105). Such losses belong to λ_{other} in (109).

7.3 Exact scope of the canonical hypothesis

The proof applies unconditionally to the canonical terms produced by direct finite Fourier decomposition of the stated fixed residue masks and by real-axis Mellin inversion of fixed smooth shapes. It does not cover any of the following broadenings without a new argument:

- a residue modulus growing with X ;
- an X -dependent smooth family whose nonzero support degenerates to a vanishing fraction of a dyadic row cell;
- a hard-cutoff boundary block failing (64), unless it is recellized and its fixed-shape or aggregate energy hypotheses are verified separately;
- a Mellin contour shift that introduces non-unit power weights; or
- an arbitrary bounded row coefficient substituted for the fixed character/phase provenance.

These exclusions are quantitative, not cosmetic: the construction in [proposition 5.6](#) gives zero survival under the last broadening.

Accordingly, the theorem neither breaches the parity barrier nor proves any fixed-shift prime-pair or twin-prime assertion. Its contribution is the sharper localization of the remaining gap: after the actual fixed-complexity provenance is unpacked before the static mask, the identified direct active-cell weighted-survival subdoor is open, while the Mellin-frame, coherent-prime, residual-grouping, and physical-localization doors remain.

8 Deterministic certificate and reproducibility boundary

The repository contains the exact-arithmetic script

`experiments/tpc50_certificate.py.`

It produces `experiments/tpc50_certificate.json`. The script uses integers and rational numbers only: there is no floating-point arithmetic, pseudorandom input, numerical optimizer, or tolerance-based comparison.

The certificate performs the following deterministic checks.

- (i) It enumerates every binary mask of size at most 3×3 in the declared suite and all nonzero integer row weights with entries in $\{0, 1, 2\}$. It verifies the exact convex-combination identity

$$\delta_w(M; a, b) = \sum_i \frac{|a_i|^2}{\|a\|_2^2} s_b(i), \quad (111)$$

and checks $\inf_{a \neq 0} \delta_w(M; a, b) = 1 - \eta_M(b)$. A basis vector at a minimizing row attains the value in every one of the 15,884 mask-right-weight cases.

- (ii) It exhausts all triples of 2×2 binary forbidden-channel matrices and verifies both the three-channel union bound for η_M and the resulting survival inequality for every enumerated left and right weight.
- (iii) It checks the degree-flatness inequality exactly and verifies the energetic-cell lower bound and the aggregate nonenergetic energy estimate for 132,864 finite selection cases at $\tau \in \{1/4, 1/2, 3/4\}$.
- (iv) It verifies invariance under independent nonzero integer scaling and under row, column, and supported-kernel phases in the exact Gaussian unit group $\{1, -1, i, -i\}$.
- (v) It constructs one explicit finite row universe containing a same-source block, a near-product block, and a large-gcd block. Each block has zero survival. Adding one admitted neighbor gives the exact positive value $1/(|S| + 1) = 1/3$ in each channel.
- (vi) It checks the rational endpoint ledger, including

$$\frac{99979}{210000} + \frac{10049}{52500} = \frac{267}{400}, \quad \frac{267}{400} - \frac{1}{400} = \frac{133}{200}, \quad (112)$$

the interior-cutoff margin $23/120 - 10049/52500 = 9/35000$, the canonical degree-defect exponent when $\kappa_0 = 1/400$, and the zero-cost isolated-face budget identity.

The committed record contains 1,469,068 exact checks. Its release digests are

semantic	b01a3c4ede8b20090e904d7f93cd29174f2d8bb46b294f9fc02d988e398786cf
source	8ad51dbe7dd900c993a616968111077ef9d55485e2d457239d314d31ee908915
JSON	f4f7cd5abb63a8919f7594ecee2c23d189ea07109906f069aab2112411d9393b

The semantic digest is taken from the canonical JSON serialization of the result payload before hash metadata are added. The source digest normalizes line endings and trailing whitespace. The JSON digest is the SHA-256 hash of the committed generated record.

This computation is a finite regression certificate, not an asymptotic proof. In particular, it does not prove the fixed-modulus prime number theorem, the squarefree progression asymptotic, the complete-universe degree estimate, or the physical provenance of a residue/Mellin cell. It does not identify the abstract obstruction weights with literal TPC coefficients, estimate the coherent prime square, or test any prime-pair or twin-prime assertion. Those logical boundaries are part of the theorem statements, not empirical caveats.

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